

MOORED BUOY 21002, Japan

WMO: 992260

Lat: 37.700N

Lon: 134.300E

Elev: 0

StdP: 14.70

Time Zone: 9.00 (E09)

Period: 90-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	28.0	30.2	20.4	15.3	37.3	22.6	17.0	37.5	49.8	35.6	46.3	38.9	27.6	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.8	82.5	78.8	81.3	77.8	80.2	76.8	79.7	81.5	78.6	81.0	77.6	80.0	13.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	150.8	81.2	77.9	144.8	80.6	76.7	139.1	79.8	43.2	81.7	42.2	81.4	41.2	80.0	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
43.8	37.3	32.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.9	39.9	39.0	43.6	50.4	57.6	64.6	72.1	77.8	72.5	64.0	54.6	45.8	(6)
	DBStd	13.62	5.47	5.66	4.81	4.40	3.72	3.04	3.73	2.63	3.61	4.55	6.09	5.92	(7)
	HDD50	1061	316	310	202	45	1	0	0	0	0	0	30	157	(8)
	HDD65	3861	779	729	662	438	230	42	2	0	1	71	312	595	(9)
	CDD50	3590	1	1	5	57	237	438	686	861	674	435	168	27	(10)
	CDD65	913	0	0	0	0	1	29	222	396	225	40	0	0	(11)
	CDH74	4235	0	0	0	0	0	0	595	2972	659	9	0	0	(12)
	CDH80	277	0	0	0	0	0	0	6	249	22	0	0	0	(13)
(14) Wind	WSAvg	15.8	20.3	19.4	16.0	14.3	12.6	11.5	12.9	13.3	15.2	16.0	18.0	20.2	(14)
(15) Precipitation	PrecAvg	40.2	1.9	1.7	1.9	3.0	3.5	3.7	5.2	5.2	5.9	3.6	2.7	2.0	(15)
	PrecMax	53.9	4.1	3.2	3.8	5.5	7.2	8.9	15.2	15.6	10.3	6.6	5.7	4.4	(16)
	PrecMin	28.0	0.8	0.4	0.2	0.4	1.1	0.9	0.6	1.5	1.4	0.7	1.0	0.5	(17)
	PrecStd	5.7	0.7	0.7	0.9	1.5	1.4	2.0	3.4	3.0	2.4	1.4	1.1	0.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	56.6	54.6	60.7	67.1	73.3	80.9	84.5	81.9	75.1	66.9	61.1	(19)
		MCWB	49.7	54.1	51.3	58.3	62.9	70.0	76.6	79.0	77.3	72.4	60.8	56.7	(20)
	2%	DB	52.6	51.9	53.4	58.9	65.4	72.0	79.5	83.4	80.6	72.3	65.5	59.5	(21)
		MCWB	47.5	48.4	50.3	56.1	62.0	69.3	76.0	79.0	76.6	67.2	59.6	54.4	(22)
	5%	DB	50.2	49.7	52.1	57.3	63.6	70.2	78.4	82.4	79.1	70.9	64.3	57.2	(23)
		MCWB	45.1	45.8	48.5	54.8	61.0	67.7	75.4	78.9	73.9	64.9	58.5	50.9	(24)
	10%	DB	48.0	47.6	50.9	55.8	62.5	68.9	77.1	81.4	77.4	70.0	62.9	54.9	(25)
		MCWB	43.0	43.2	47.1	53.1	60.0	66.5	74.4	78.1	72.8	63.6	57.1	48.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.4	56.2	52.2	59.5	64.5	71.0	78.2	81.0	79.6	73.5	62.8	59.3	(27)
		MCDB	55.0	58.0	53.8	60.5	65.7	72.5	79.9	82.1	80.7	75.7	64.9	60.8	(28)
	2%	WB	48.4	49.9	51.1	56.7	63.0	69.8	77.0	80.2	77.7	69.9	61.6	54.1	(29)
		MCDB	52.3	51.3	53.1	58.3	64.1	72.0	78.9	81.8	79.9	71.7	64.3	58.8	(30)
	5%	WB	45.8	46.5	49.5	55.4	61.8	68.2	76.1	79.5	75.8	67.7	59.7	51.2	(31)
		MCDB	50.0	49.7	51.7	57.2	63.1	69.8	78.0	81.5	78.1	70.0	63.2	55.8	(32)
	10%	WB	43.8	43.3	48.0	53.9	60.5	67.1	75.1	78.7	74.5	65.3	57.7	48.9	(33)
		MCDB	47.9	47.5	50.7	55.7	62.1	68.7	77.1	81.2	77.0	69.4	62.2	54.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	6.3	5.9	4.3	3.7	2.9	2.7	2.8	3.3	4.8	6.6	7.2	(35)
		MCDBR	10.3	8.6	7.2	4.1	3.8	2.9	3.2	3.2	4.0	5.4	8.5	8.5	(36)
	5% WB	MCWBR	10.3	8.9	8.3	4.2	3.6	3.1	2.6	3.0	4.8	7.3	9.3	9.0	(37)
		MCWBR	10.7	7.5	7.1	3.7	4.2	2.8	2.7	3.6	6.5	7.8	9.3	9.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.431	0.462	0.526	0.553	0.556	0.555	0.562	0.509	0.457	0.446	0.437	0.437	(40)
		taud	2.012	1.946	1.825	1.812	1.875	1.962	2.011	2.178	2.265	2.141	2.081	1.996	(41)
	Ebn at Noon		224	235	232	236	238	238	235	245	252	239	221	210	(42)
		Edh at Noon	41	50	62	66	63	58	55	46	40	41	38	39	(43)
(44) All-Sky Solar Radiation	RadAvg		492	794	1242	1610	1735	1707	1589	1673	1339	1054	641	435	(44)
	RadStd		52	73	92	137	166	130	180	182	109	86	62	43	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page